

HW2-F25-soln

September 16, 2025

1 Advanced Chemical Reaction Engineering (CBE 60546)

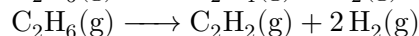
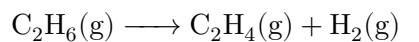
1.1 Homework 2

1.2 Due September 12, 2025

Carefully and neatly document your answers. You may use a mathematical solver like Jupyter/iPython. Use plotting software for all plots.

2 Shale, shale everywhere

2.1 Your nation has stumbled upon an enormous reserver of ethane (that's really true!), which you propose to use as a feedstock to produce ethylene, acetylene, and hydrogen via thermal dehydrogenation:



2.2 The thermodynamic data [here](#) is from the [CRC Handbook](#), and is based on 1 bar standard state. All the gases can be considered to be ideal over the range of interest here.

```
[1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy.optimize import minimize
from scipy.optimize import root
from scipy.optimize import curve_fit
# from google.colab import files # This is required only when you are using
↳ Colab.
```

2.2.1 What does CSV stand for?

- CSV stands for **Comma-Separated Values**.
- It is a simple text file format where each line corresponds to a row of data, and values in each row are separated by commas.

2.2.2 Why is the pandas package good for reading CSV files?

- The **pandas** library in Python provides the powerful function `pd.read_csv()` that can:
- Load CSV data directly into a **DataFrame** (a 2D table-like structure with labeled rows and columns).
- Automatically parse headers (column names).
- Handle missing values gracefully.
- Read large files efficiently, with options for chunking.
- Support advanced features like specifying delimiters (`,`, `;`, `\t`), selecting only certain columns, parsing dates, etc.

```
[2]: # Upload HW2-thermo.csv with colab.files, read it and store it as a Pandas
      ↪ dataframe.
      # If you use Jupyter on you own laptop, you don't need to upload the file.
      df = pd.read_csv("HW2-thermo.csv")
      df.head() # This displays top 5 rows
```

```
[2]:
```

	Row		Subheader	T (K)	Cp° (J/mol K)	\
0	156	12. ACETYLENE	C₂H₂ (g)	298.15	44.036	
1	157	12. ACETYLENE	C₂H₂ (g)	300.00	44.174	
2	158	12. ACETYLENE	C₂H₂ (g)	400.00	50.388	
3	159	12. ACETYLENE	C₂H₂ (g)	500.00	54.751	
4	160	12. ACETYLENE	C₂H₂ (g)	600.00	58.121	

	S° (J/mol K)	-(G°-H°(T)) (J/mol K)	(H-H°(T)) (kJ/mo)	Hf° (kJ/mol)	\
0	200.927	200.927	0.000	227.400	
1	201.199	200.927	0.082	227.397	
2	214.814	202.741	4.829	227.161	
3	226.552	206.357	10.097	226.846	
4	236.842	210.598	15.747	226.445	

	Gf° (kJ/mol)	Log Kf
0	209.879	-36.769
1	209.770	-36.524
2	203.928	-26.630
3	198.154	-20.701
4	192.452	-16.754

```
[3]: df.tail() # This displays bottom 5 rows
```

```
[3]:
```

	Row		Subheader	T (K)	Cp° (J/mol K)	\
51	702	51. DIHYDROGEN	H₂ (g)	1100.0	30.580	
52	703	51. DIHYDROGEN	H₂ (g)	1200.0	30.991	
53	704	51. DIHYDROGEN	H₂ (g)	1300.0	31.422	
54	705	51. DIHYDROGEN	H₂ (g)	1400.0	31.860	
55	706	51. DIHYDROGEN	H₂ (g)	1500.0	32.296	

	S° (J/mol K)	-(G°-H°(T)) (J/mol K)	(H-H°(T)) (kJ/mo)	Hf° (kJ/mol)	\
--	--------------	-----------------------	-------------------	--------------	---

51	169.113	147.550	23.719	0.0
52	171.791	149.460	26.797	0.0
53	174.288	151.275	29.918	0.0
54	176.633	153.003	33.082	0.0
55	178.846	154.653	36.290	0.0

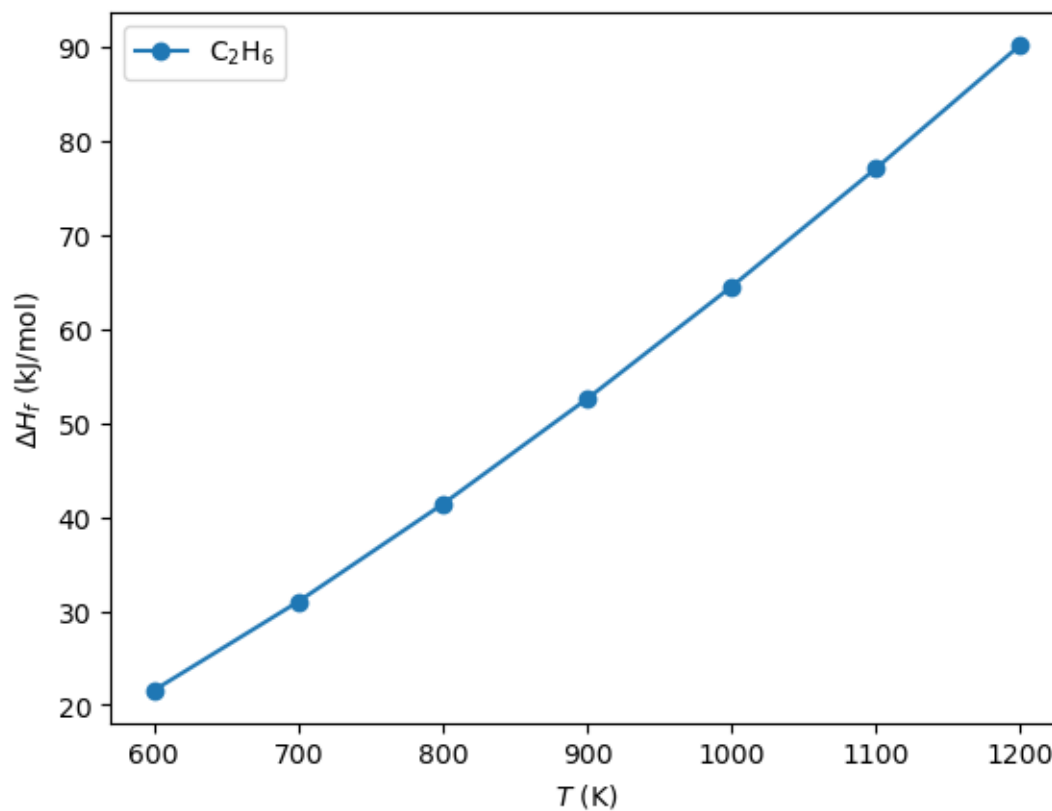
	Gf° (kJ/mol)	Log Kf
51	0.0	0.0
52	0.0	0.0
53	0.0	0.0
54	0.0	0.0
55	0.0	0.0

2.2.3 Q1.1. Plot the standard state enthalpy of ethane vs T from 600 to 1200 K. What does the slope of the plot correspond to?

```
[4]: # Find and search indexes for ETHANE data by name in 'Subheader' column
idx = np.where(df.Subheader.str.contains("ETHANE", case=False, na=False))[0]
df_ethane = df.iloc[idx[0]:idx[-1]]
# Filter data by temperature range
df_ethane = df_ethane[(df_ethane['T (K)'] >= 600) & (df_ethane['T (K)'] <=
↳ 1200)]
# Show top 5 rows for ETHANE data
print(df_ethane.head())
# Standard state enthalpy of ethane corresponds to '(H-H°(T)) (kJ/mo)' not
↳ 'Hf° (kJ/mol)'
plt.plot(df_ethane['T (K)'], df_ethane['(H-H°(T)) (kJ/mo)'], 'o-', label =
↳ 'C$_2$H$_6$')
plt.legend()
plt.xlabel('$T$ (K)')
plt.ylabel(r'$H_f$ (kJ/mol)')
plt.show()
```

	Row		Subheader	T (K)	Cp° (J/mol K)	\
32	188	14.	ETHANE C ₂ H ₆ (g)	600.0	89.188	
33	189	14.	ETHANE C ₂ H ₆ (g)	700.0	99.136	
34	190	14.	ETHANE C ₂ H ₆ (g)	800.0	107.936	
35	191	14.	ETHANE C ₂ H ₆ (g)	900.0	115.709	
36	192	14.	ETHANE C ₂ H ₆ (g)	1000.0	122.552	
	S° (J/mol K)		-(G°-H°(T)) (J/mol K)	(H-H°(T)) (kJ/mo)	Hf° (kJ/mol)	\
32	277.568		241.660	21.545	-96.779	
33	292.080		247.835	30.972	-99.663	
34	305.904		254.236	41.334	-101.963	
35	319.075		260.715	52.525	-103.754	
36	331.628		267.183	64.445	-105.105	

	Gf° (kJ/mol)	Log Kf
32	26.086	-2.271
33	46.800	-3.492
34	67.887	-4.433
35	89.231	-5.179
36	110.750	-5.785



2.2.4 Q1.2. Is the slope positive or negative? Why?

For the “standard enthalpy” you plotted as the sensible enthalpy, $H(T) - H^\circ(298 \text{ K})$,

$$H(T) - H^\circ(298) = \int_{298}^T C_p(T') dT'$$

so the slope of the curve is

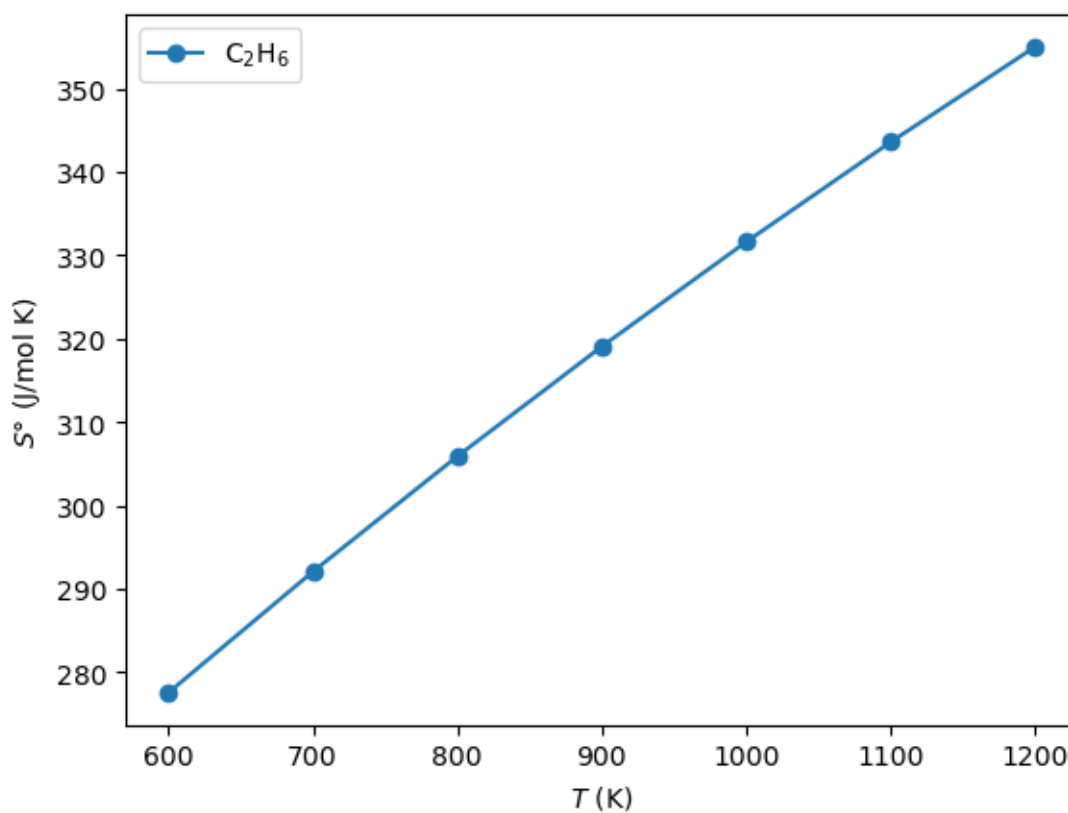
$$\frac{d}{dT} [H(T) - H^\circ(298)] = C_p(T).$$

Therefore, the plot has a **positive slope**, and its numerical value at any T equals the molar heat capacity $C_p(T)$ (in $\text{kJ mol}^{-1} \text{K}^{-1}$).

2.2.5 Q1.3. Plot the standard state entropy of ethane vs T from 600 to 1200 K.

```
[5]: # Answer here
plt.plot(df_ethane['T (K)'], df_ethane['S° (J/mol K)'], 'o-', label = 'C2H6')
# plt.xlim(600,1200)
# plt.ylim(275,360)
plt.legend()
plt.xlabel('T (K)')
plt.ylabel('S° (J/mol K)')
plt.show()

print('The slope corresponds to Cv/T.')
```

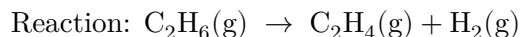


The slope corresponds to C_v/T .

2.2.6 Q1.4. Is the slope positive or negative? Why?

Entropy S° increases with temperature because $\left(\frac{\partial S}{\partial T}\right)_P = \frac{C_p}{T}$, and since $C_p > 0$ and $T > 0$, the slope is always positive.

2.2.7 Q1.5. Consider the ethane to ethylene reaction. What do you expect for the sign of ΔH° , ΔS° , and ΔG° ? Why?



Signs & Why

- **Enthalpy, $\Delta H^\circ > 0$ (endothermic).**
Breaking C–H single bonds and making a C=C plus H–H typically requires net energy input.
- **Entropy, $\Delta S^\circ > 0$.**
One gas molecule becomes **two** gas molecules, so disorder and the number of accessible microstates increase.
- **Gibbs free energy, $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \rightarrow$ temperature-dependent sign.**
With $\Delta H^\circ > 0$ and $\Delta S^\circ > 0$:
 - At **low** T , the positive ΔH° dominates $\Delta G^\circ > 0$ (not spontaneous).
 - At **high** T , the $T\Delta S^\circ$ term can outweigh ΔH° $\Delta G^\circ < 0$ (favorable).

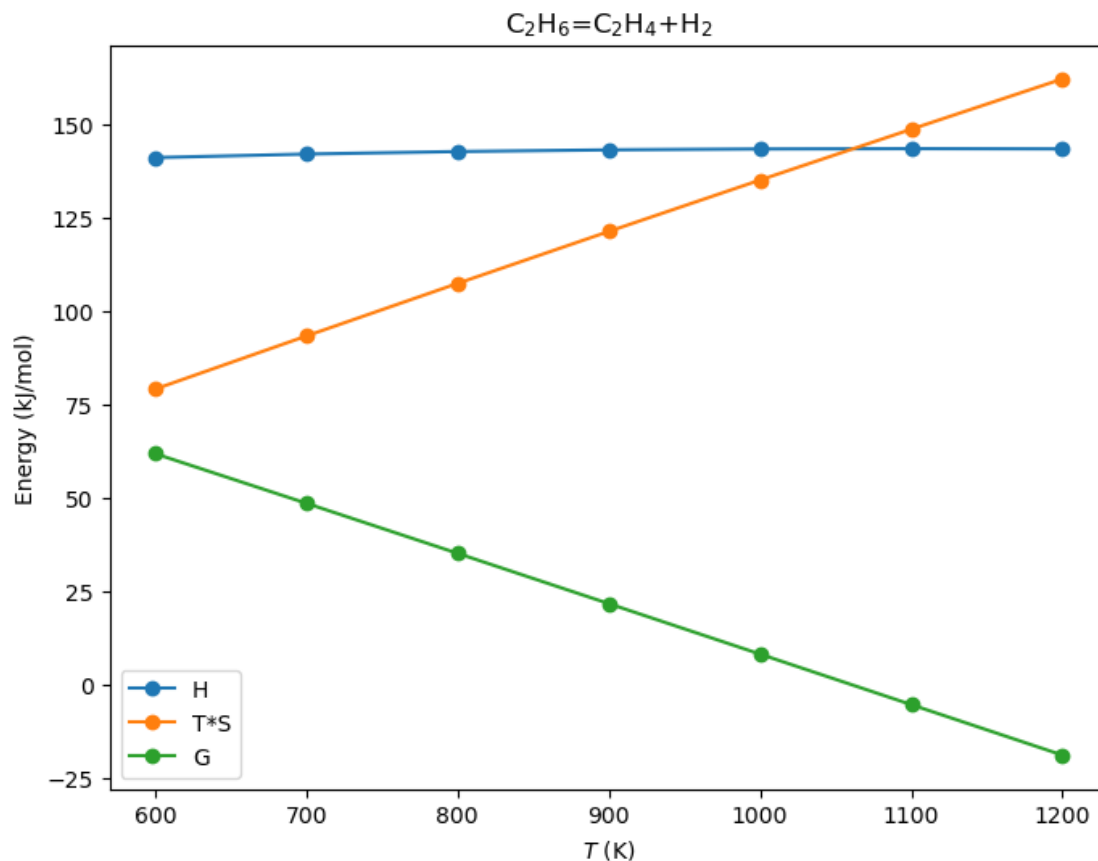
2.2.8 Q1.6. Plot the standard state reaction enthalpy, T times the standard state reaction entropy, and the standard state reaction Gibbs energy vs T from 600 to 1200 K.

```
[6]: # Answer Here
idx = np.where(df.Subheader.str.contains("ETHYLENE", case=False, na=False))[0]
df_C2H4 = df.iloc[idx[0]:idx[-1]]
df_C2H4 = df_C2H4[(df_C2H4['T (K)'] >= 600) & (df_C2H4['T (K)'] <= 1200)]
idx = np.where(df.Subheader.str.contains("DIHYDROGEN", case=False, na=False))[0]
df_H2 = df.iloc[idx[0]:idx[-1]]
df_H2 = df_H2[(df_H2['T (K)'] >= 600) & (df_H2['T (K)'] <= 1200)]

# Standard state reaction enthalpy vs T, C2H6 = C2H4 + H2
plt.figure(figsize=(8,6))
dH_C2H6_C2H4 = df_H2['Hf° (kJ/mol)'].values + df_C2H4['Hf° (kJ/mol)'].values -
    df_ethane['Hf° (kJ/mol)'].values
plt.plot(df_ethane['T (K)'], dH_C2H6_C2H4, 'o-', label = 'H')
dS_C2H6_C2H4 = df_H2['S° (J/mol K)'].values + df_C2H4['S° (J/mol K)'].values -
    df_ethane['S° (J/mol K)'].values
plt.plot(df_ethane['T (K)'], dS_C2H6_C2H4*df_ethane['T (K)'].values/1000, 'o-',
    label = 'T*S')
dG_C2H6_C2H4 = dH_C2H6_C2H4 - df_ethane['T (K)'].values * dS_C2H6_C2H4 / 1000
plt.plot(df_ethane['T (K)'], dG_C2H6_C2H4, 'o-', label = 'G')

# plt.xlim(600,1200)
# plt.ylim(140,)
plt.title('C$2$H$6$=C$2$H$4$+H$2$')
plt.legend()
```

```
plt.xlabel('$T$ (K)')
plt.ylabel('Energy (kJ/mol)')
plt.show()
```



2.2.9 Q1.7. What do the plots tell you about the conditions necessary to run this reaction thermally? About what contributes to the free energy change with T ?

```
[7]: # Answer Here
# Get T with delta G = 0 with linear interpolation
# np.interp only works for monotonically increasing sample points. dG_C2H6_C2H4_
    ↪ was decreasing though.
T_G0 = np.interp(0,dG_C2H6_C2H4[::-1],df_ethane['T (K)'].values[::-1])
print ('Standard state delta G = 0 at {:.2f} K.'.format(T_G0))
print ('Plot tells us that increasing temperature favors products; ie,
    ↪ equilibrium extent of reaction increases with T.')
print ('A positive reaction entropy contributes to the free energy change with
    ↪ T.')
```

Standard state delta G = 0 at 1061.59 K.

Plot tells us that increasing temperature favors products; ie, equilibrium

extent of reaction increases with T.

A positive reaction entropy contributes to the free energy change with T.

2.2.10 Q1.8. Plot the free energy of the ethane to ethylene reaction vs ethane conversion at 800, 1000, and 1200K and 1 bar. Determine the equilibrium conversion of ethane at each temperature if this was the only reaction to occur.

$$G(T, n_j) = \sum_j \mu(T)_j n_j$$

$$\mu_j(T) = \mu_j^\circ(T) + RT(\ln y_j + \ln P)$$

$$G(T, n_j) = \sum_j \mu_j^\circ(T) n_j + RT \sum_j n_j (\ln y_j + \ln P)$$

$$n_j = n_{j0} + \nu_j \xi; y_j = n_j / \sum n_j$$

$$G(T, \xi) = \sum_j \mu_j^\circ(T) n_{j0} + \xi \sum_j \nu_j \mu_j^\circ(T) + RT \sum_j N_j (\ln y_j + \ln P)$$

Normalize G by $n_{\text{C}_2\text{H}_6,0}$. Let $\xi' = \xi / n_{\text{C}_2\text{H}_6,0}$ be a dimensionless advancement.

$$G(T, \xi') / \text{mol C}_2\text{H}_6 = \sum_j \mu_j^\circ(T) y_{j0} + \xi' \sum_j \nu_j \mu_j^\circ + RT \sum_j \frac{n_j}{n_{\text{C}_2\text{H}_6,0}} (\ln y_j + \ln P)$$

$$G(T, \xi') = \sum_j \mu_j^\circ(T) y_{j0} + \xi' \Delta G_r^\circ(T) + RT \sum_j \frac{n_j}{n_{\text{C}_2\text{H}_6,0}} (\ln y_j + \ln P)$$

$$G(T, \xi') = \sum_j \mu_j^\circ(T) y_{j0} + \xi' \Delta G_r^\circ(T) + RT \sum_j (y_{j0} + \nu_j \xi') (\ln y_j + \ln P)$$

Take $y_{\text{ethane},0} = 1$, $y_{\text{ethylene},0} = y_{\text{H}_2,0} = 0$

$$G(T, \xi') = \mu_{\text{C}_2\text{H}_6}^\circ(T) + \xi' \Delta G_r^\circ(T) + RT \sum_j (y_{j0} + \nu_j \xi') (\ln y_j + \ln P)$$

This G has a T dependent intercept that is the standard state free energy of ethane at the given T . That can be figured out from the Table above and is a decreasing function of T . For ease of visualization, one can subtract off this reference energy and normalize to RT to have all plots start at 0.

$$\bar{G} = (G - \mu_{\text{C}_2\text{H}_6}^\circ) / RT = \xi' \Delta G_r^\circ / RT + \sum_j (y_{j0} + \nu_j \xi') (\ln y_j + \ln P)$$

This is what is plotted below.

```
[ ]:
[1]: # Reaction: C2H6 = C2H4 + H2
# First define the free energy (see the note) as a function of dimensionless
# advancement
# G_bar = (G(xi,T,P) - y_C2H6_0*mu_C2H6_0)/RT = ksi*deltaG/R/T + (y_H2_0 +
# nu_H2*ksi)(lny_H2 + lnP) + (y_C2H4_0 + nu_C2H4*ksi)(lny_C2H4 + lnP) +
# (y_C2H6_0 - nu_C2H6*ksi)(lny_C2H6 + lnP)
# G is the free energy of the system. ksi is the dimensionless advancement.
# deltaG is the standard free energy, y is the fraction, nu is the
# stoichiometric coefficient and P is the total pressure.
def G_C2H6_C2H4_H2(ksi):
    global y_H2_0,y_C2H4_0,y_C2H6_0,nu_H2,nu_C2H4,nu_C2H6,P,deltaG,T,R
```



```

y_H2 = (y_H2_0+ksi)/(1+ksi)
y_C2H4 = (y_C2H4_0+ksi)/(1+ksi)
y_C2H6 = (y_C2H6_0-ksi)/(1+ksi)
G0xi = ksi*deltaG/R/T*1000
Gxi_H2 = (y_H2_0 + nu_H2*ksi)*(np.log(y_H2) + np.log(P))
Gxi_C2H4 = (y_C2H4_0 + nu_C2H4*ksi)*(np.log(y_C2H4) + np.log(P))
Gxi_C2H6 = (y_C2H6_0 + nu_C2H6*ksi)*(np.log(y_C2H6) + np.log(P))
G_bar = G0xi+Gxi_H2+Gxi_C2H4+Gxi_C2H6
G = G_bar*R*T + df_ethane[df_ethane['T (K)']==T]['Gf° (kJ/mol)'].iloc[0]*1000
return G_bar

```

[9]: *# Define y, P and nu for this reaction*

```

R = 8.314
y_H2_0 = 0
y_C2H4_0 = 0
y_C2H6_0 = 1
P = 1
nu_H2 = 1
nu_C2H4 = 1
nu_C2H6 = -1

```

[10]: *# First define an array of dimensionless advancement ksi.*

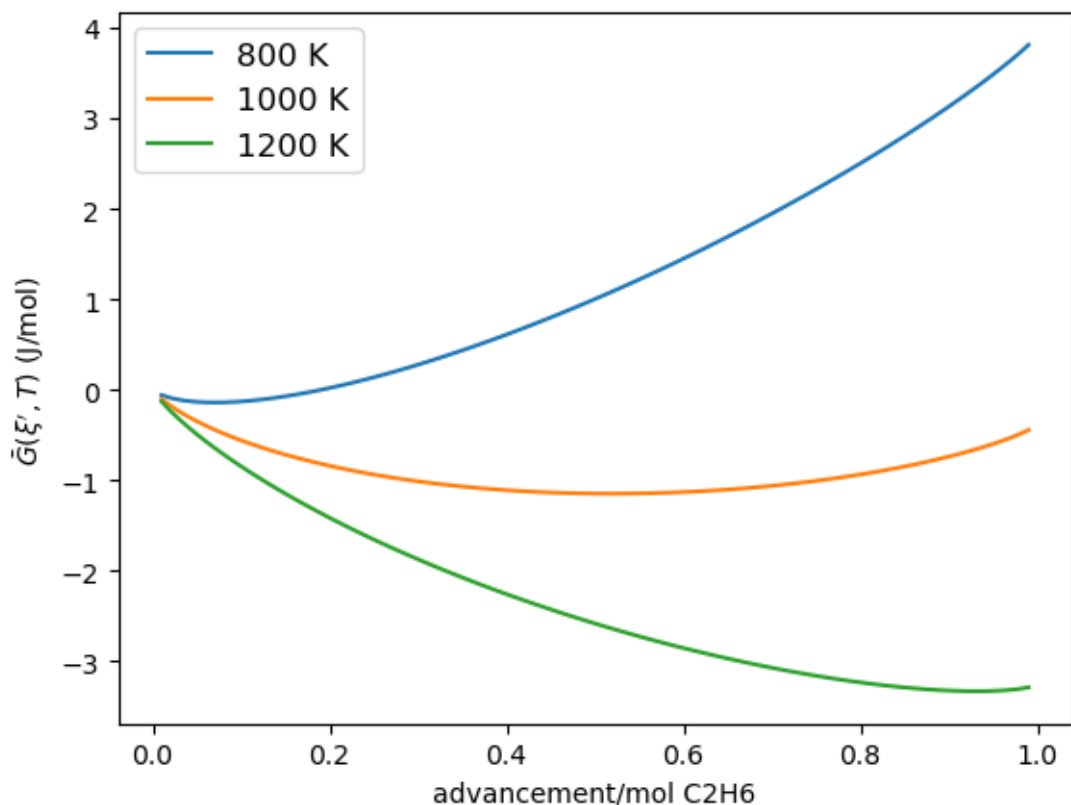
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# N_C2H6 = N_C2H6_0 - N_C2H6_0*ksi. Conversion = (N_C2H6_0-N_C2H6)/N_C2H6_0 =  $\xi$ 
↳ ksi
ksi_array = np.arange(0.01,1,0.01)
conversion_array = ksi_array
T=800
deltaG = dG_C2H6_C2H4[df_ethane['T (K)'] == T][0] # Grab deltaG reaction from  $\xi$ 
↳ above
dG_reaction = G_C2H6_C2H4_H2(ksi_array) # compute advancement part
plt.plot(conversion_array, dG_reaction, '-', label = '800 K')

T=1000
deltaG = dG_C2H6_C2H4[df_ethane['T (K)'] == T][0]
dG_reaction = G_C2H6_C2H4_H2(ksi_array)
plt.plot(conversion_array, dG_reaction, '-', label = '1000 K')

T=1200
deltaG = dG_C2H6_C2H4[df_ethane['T (K)'] == T][0]
dG_reaction = G_C2H6_C2H4_H2(ksi_array)
plt.plot(conversion_array, dG_reaction, '-', label = '1200 K')
# plt.xlim(600,1200)
# plt.ylim(-25,65)
plt.legend(fontsize=12)
plt.xlabel('advancement/mol C2H6')
plt.ylabel(r'$\bar{G}(\xi^{\prime},T)$ (J/mol)')
plt.show()

```



2.2.11 Q1.9. Imagine a semi-permeable membrane through which you could control the partial pressure (chemical potential) of hydrogen in a reaction vessel. Plot the mole fractions of ethane and ethylene as a function of $\ln P_{H_2}$ at 1000K. (Assume the H_2 does not contribute to the total moles in the vessel.)

Reaction vessel conditions

In a reaction vessel, the total volume is constant.

For this reaction, the total pressure is also constant:

$$P_{C_2H_6} + P_{C_2H_4} = P_{C_2H_6}^0$$

Dividing through by

$$P_{C_2H_6}^0$$

:

$$\frac{P_{C_2H_6}}{P_{C_2H_6}^0} + \frac{P_{C_2H_4}}{P_{C_2H_6}^0} = 1$$

Define mole fractions:

$$y_{C_2H_6} + y_{C_2H_4} = 1, \quad y_{C_2H_4} = 1 - y_{C_2H_6}$$

Equilibrium constant:

$$K_{eq} = \frac{\left(\frac{P_{H_2}}{P^0}\right) \left(\frac{P_{C_2H_4}}{P^0}\right)}{\frac{P_{C_2H_6}}{P^0}} = \frac{P_{H_2}}{P^0} \cdot \frac{P_{C_2H_4}/P_{C_2H_6}^0}{P_{C_2H_6}/P_{C_2H_6}^0}$$

So:

$$K_{eq} = \frac{P_{H_2}}{P^0} \cdot \frac{y_{C_2H_4}}{y_{C_2H_6}}$$

Substitute

$$y_{C_2H_4} = 1 - y_{C_2H_6}$$

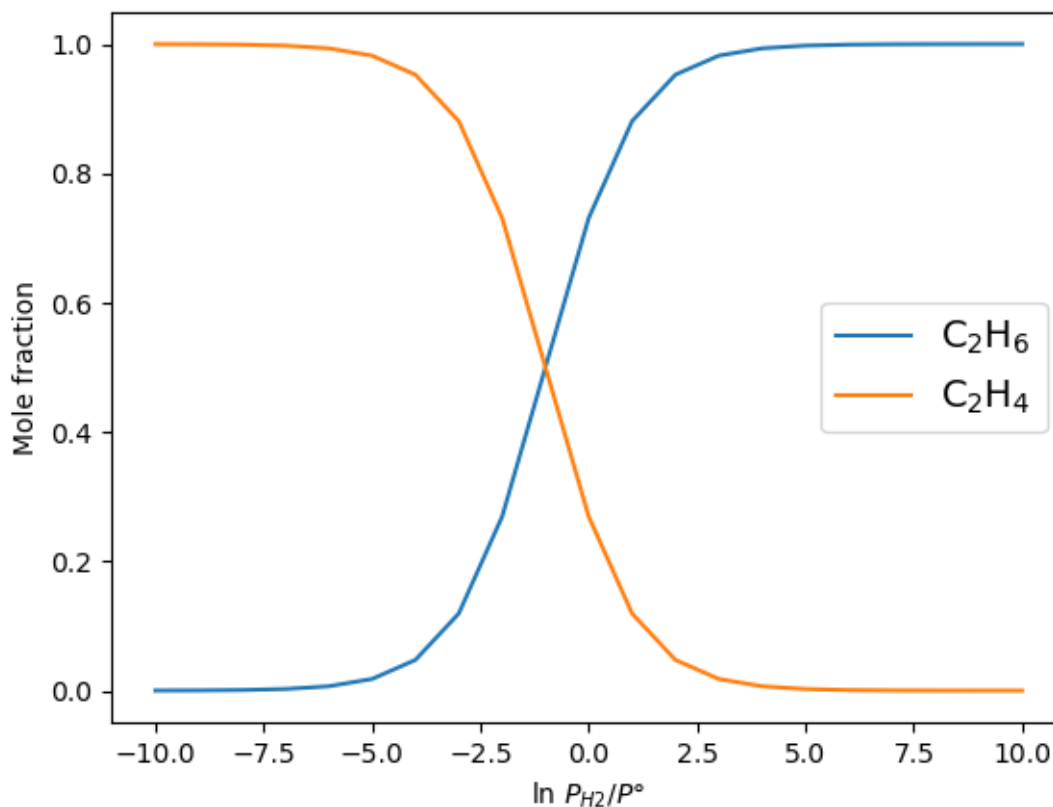
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$$K_{eq} = \frac{P_{H_2}}{P^0} \cdot \frac{1 - y_{C_2H_6}}{y_{C_2H_6}}$$

Final expression:

$$y_{C_2H_6} = \frac{\frac{P_{H_2}}{P^0}}{\frac{P_{H_2}}{P^0} + K_{eq}}$$

```
[11]: T=1000
deltaG = dG_C2H6_C2H4[df_ethane['T (K)'] == T][0]*1000 # kj/mol to j/mol
K_eq = np.exp(-deltaG/R/T)
ln_P_array = np.arange(-10,11,1)
P_H2_array = np.exp(ln_P_array)
y_C2H6_array = P_H2_array / (P_H2_array+K_eq)
plt.plot(ln_P_array,y_C2H6_array, '-', label = 'C$_2$H$_6$')
plt.plot(ln_P_array,1-y_C2H6_array, '-', label = 'C$_2$H$_4$')
plt.legend(fontsize=14)
plt.xlabel(r'\ln $P_{H_2}$/$P^\circ$')
plt.ylabel(r'Mole fraction')
plt.show()
```



2.2.12 Q1.10. Now consider both reactions simultaneously. Determine the equilibrium ratio of ethylene to acetylene, starting from pure ethane, at 1000 K and 1, 10, and 100 bar total pressure (again, assume ideal behavior at all conditions). Can you explain the result?

	C_2H_6	C_2H_4	H_2	C_2H_2
Initial	1	0	0	0
Change	$-\epsilon_1 - \epsilon_2$	ϵ_1	$\epsilon_1 + 2\epsilon_2$	ϵ_2
End	$1 - \epsilon_1 - \epsilon_2$	ϵ_1	$\epsilon_1 + 2\epsilon_2$	ϵ_2

$$total = 1 + \epsilon_1 + 2\epsilon_2$$

Two equations:

$$K_{eq1} = \frac{y_{C_2H_4} y_{H_2}}{y_{C_2H_6}} \left(\frac{P}{P^o} \right) = \frac{\frac{\epsilon_1}{1+\epsilon_1+2\epsilon_2} \frac{\epsilon_1+2\epsilon_2}{1+\epsilon_1+2\epsilon_2}}{\frac{1-\epsilon_1-\epsilon_2}{1+\epsilon_1+2\epsilon_2}} \left(\frac{P}{P^o} \right) = \frac{\epsilon_1(\epsilon_1+2\epsilon_2)}{(1+\epsilon_1+2\epsilon_2)(1-\epsilon_1-\epsilon_2)} \left(\frac{P}{P^o} \right)$$

$$K_{eq2} = \frac{y_{C_2H_2} y_{H_2}^2}{y_{C_2H_6}} \left(\frac{P}{P^o} \right)^2 = \frac{\epsilon_2(\epsilon_1+2\epsilon_2)^2}{(1+\epsilon_1+2\epsilon_2)^2(1-\epsilon_1-\epsilon_2)} \left(\frac{P}{P^o} \right)^2$$

solve for ϵ_1 and ϵ_2

```
[12]: # Calculate  $G_f^\circ$  for the first reaction

ethane_G = df['  $G_f^\circ$  (kJ/mol)'][32:39]
hydrogen_G = df['  $G_f^\circ$  (kJ/mol)'][46:53]
ethylene_G = df['  $G_f^\circ$  (kJ/mol)'][18:25]

dG_r1 = hydrogen_G.values + ethylene_G.values - ethane_G.values
K_r1_1000K = np.exp(-dG_r1[4]/1000/0.008314)

[13]: # Calculate  $G_f^\circ$  for the second reaction
ethane_G = df['  $G_f^\circ$  (kJ/mol)'][32:39]
hydrogen_G = df['  $G_f^\circ$  (kJ/mol)'][46:53]
acetylene_G = df['  $G_f^\circ$  (kJ/mol)'][18:25]

dG_r2_1000K = 2*df['  $G_f^\circ$  (kJ/mol)'][50] + df['  $G_f^\circ$  (kJ/mol)'][8] - df['  $G_f^\circ$  (kJ/
    ↪mol)'][36]

#Calculate  $K_{eq}$  for second reaction
K_r2_1000K = np.exp(-dG_r2_1000K/1000/0.008314)

[14]: #solve for extension of reaction
from scipy.optimize import fsolve
# print(c1,c2)
def equations(p):
    c1 = K_r1_1000K/P
    c2 = K_r2_1000K/P**2
    e1,e2 = p
    ↪
    ↪return(e1*(e1+2*e2)-c1*(1+e1+2*e2)*(1-e1-e2),e2*(e1+2*e2)**2-c2*(1+e1+2*e2)**2*(1-e1-e2))
P = 1 #bar
e1,e2 = fsolve(equations, (0.5,0.5))
print('Equilibrum ratio of ethane to acethlyene at 1 bar is ',e1/e2)
print (e1,e2)
P = 10
e1,e2 = fsolve(equations, (0.5,0.5))
print('Equilibrum ratio of ethane to acethlyene at 10 bar is ',e1/e2)
print (e1,e2)

P = 100
e1,e2 = fsolve(equations, (0.5,0.5))
print('Equilibrum ratio of ethane to acethlyene at 100 bar is ',e1/e2)
print (e1,e2)

print('As pressure increase, the conversion of second reaction decrease. \nThis
    ↪is because the second reaction can produce more gas than the first reaction
    ↪with same amount of ethane, the higher the pressure the more disfavor for
    ↪second reaction. ')

```

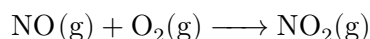
Equilibrium ratio of ethane to acetylene at 1 bar is 163.833896788789
 0.5157785342538398 0.003148179615838412
 Equilibrium ratio of ethane to acetylene at 10 bar is 757.9113580419207
 0.18810644660655823 0.00024819056293408115
 Equilibrium ratio of ethane to acetylene at 100 bar is 2727.1888957992073
 0.060509373345017166 2.2187452229004765e-05

As pressure increase, the conversion of second reaction decrease.

This is because the second reaction can produce more gas than the first reaction with same amount of ethane, the higher the pressure the more disfavor for second reaction.

3 NO_x, NO_x, who's there?

3.1 A simple reaction of NO is it's oxidation to NO₂:



3.2 You can assume all gases behave ideally under the conditions considered in this problem.

source: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C10102439&Mask=1>

3.2.1 Q2.1. Determine $\Delta H^\circ(298 \text{ K})$, $\Delta S^\circ(298 \text{ K})$, $\Delta G^\circ(298 \text{ K})$, and $K_p(298 \text{ K})$ for the NO oxidation reaction (Don't forget to balance it first!). Be sure to specify your source and the standard state.

```
[15]: # Answer here
import numpy as np
import matplotlib.pyplot as plt
R = 8.314 #[=]J/molK
T = 298 #[=]K
H_rxn = 2*33.1-2*90.29-1*0 #[=]kJ/mol
S_rxn = 2*240.04-2*210.76-1*205.152 #[=]J/mol
G_rxn = H_rxn-T*S_rxn/1000
Kp = np.exp(-G_rxn*1000/(R*T))
```

```
[16]: print(f" H(298) = {H_rxn:.2f} kJ/mol")
print(f" S(298) = {S_rxn/1000:.2f} kJ/mol*K")
print(f" G(298) = {G_rxn:.2f} kJ/mol")
print(f"Kp = {Kp:.3g}")
```

```
H(298) = -114.38 kJ/mol
S(298) = -0.15 kJ/mol*K
G(298) = -70.70 kJ/mol
Kp = 2.47e+12
```

3.2.2 Q2.2. Calculate the equilibrium partial pressure ratio of NO_2 to NO in the atmosphere near the surface of the earth. Assume the mixing ratio of O_2 to be 0.2 and a temperature of 25°C .

$$K_p^\circ = \frac{p_{\text{NO}_2}^2}{p_{\text{NO}}^2 p_{\text{O}_2}} \Rightarrow \frac{p_{\text{NO}_2}}{p_{\text{NO}}} = \sqrt{K_p^\circ p_{\text{O}_2}} = \sqrt{(2.42 \times 10^{12})(0.2)}$$

$$= 6.96 \times 10^5 = \frac{p_{\text{NO}_2}}{p_{\text{NO}}}$$

```
[17]: # Answer here
P02 = 1*0.2
P_ratio = np.sqrt(Kp*P02)
print(f'The pressure ratio is: {P_ratio:.3g}')
```

The pressure ratio is: 7.02e+05

3.2.3 Q2.3. From standard compilations and at 1 atm standard state, $\Delta H^\circ(250\text{ K}) = -116.532\text{ kJ mol}^{-1}$ and $\Delta S^\circ(250\text{ K}) = -152.179\text{ J/mol/K}$. Use the van't Hoff relationship to plot $\Delta G^\circ(T)$ vs T from room temperature to 1000°C . Add a point on your plot for the $\Delta G^\circ(298\text{ K})$ you found from a tabulation.

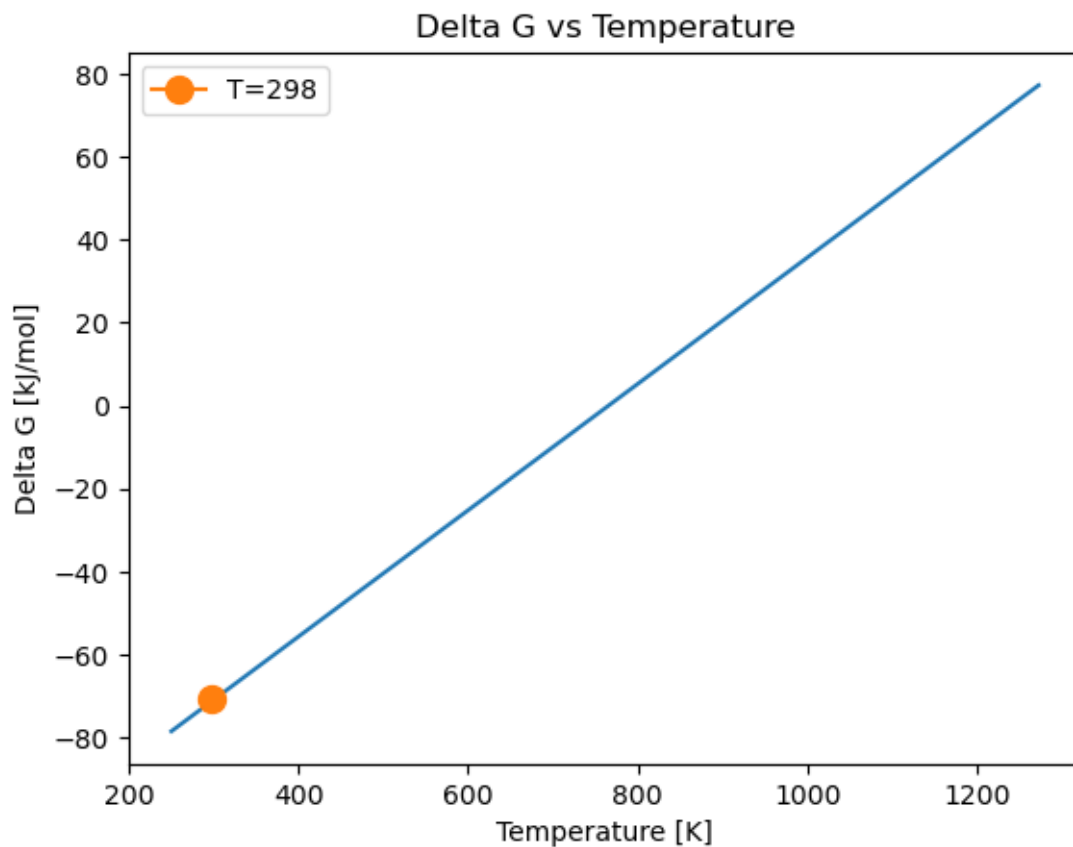
Van't Hoff:

$$\ln K_p = -\frac{\Delta G^\circ}{RT} = -\frac{\Delta H^\circ}{R} \frac{1}{T} + \frac{\Delta S^\circ}{R}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

```
[18]: # Answer here
T = np.linspace(250, 1000+273, 100)
H = -116.532 # kj/mol
S = -152.179*10**-3 # kj/mol
G = H-T*S
plt.plot(T,G)
plt.plot(298, -70.65, marker='o',markersize=10,label='T=298')
plt.xlabel('Temperature [K]')
plt.ylabel('Delta G [kJ/mol]')
plt.title('Delta G vs Temperature')
plt.legend()
```

```
[18]: <matplotlib.legend.Legend at 0x147e5bd10>
```



Van't Hoff:

$$\ln \frac{K_p(T_2)}{K_p^\circ} = -\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{298} \right)$$

$$K_p(T_2) = K_p^\circ \exp \left[-\frac{\Delta H^\circ}{R} \left(\frac{1}{T_2} - \frac{1}{298} \right) \right]$$

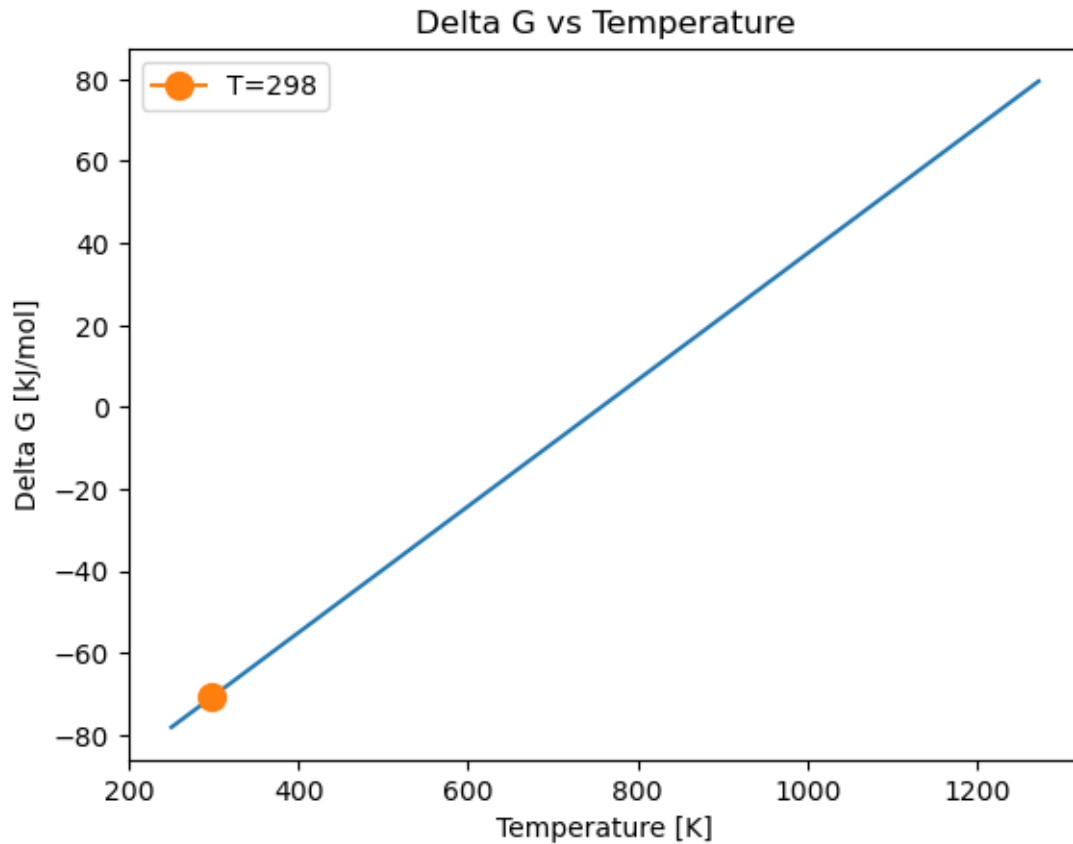
$$\Delta G(T) = -RT \ln(K_p(T))$$

```
[19]: T = np.linspace(250, 1000+273, 100)
R = 8.314 * 10**-3 # kJ/mol*K
H = -116.532 # kJ/mol
Kp0 = 2.42*10**12 # from Q3.1
Kp = Kp0*np.exp(-H/R*(1/T-1/298))
G_T = -R*T*np.log(Kp)
plt.plot(T,G_T)
plt.plot(298, -70.65, marker='o',markersize=10,label='T=298')
plt.xlabel('Temperature [K]')
```



```
plt.ylabel('Delta G [kJ/mol]')
plt.title('Delta G vs Temperature')
plt.legend()
```

[19]: <matplotlib.legend.Legend at 0x147d73dd0>



3.2.4 Q2.4. NO oxidation is catalyzed over diesel oxidation catalysts (DOCs) on diesel vehicles. Plot the equilibrium conversion of NO to NO₂ vs T from room temperature to 1000°C for an isobaric 1 atm reactor presented with 0.1% NO and 5% O₂, and balance N₂.

Conversion relation

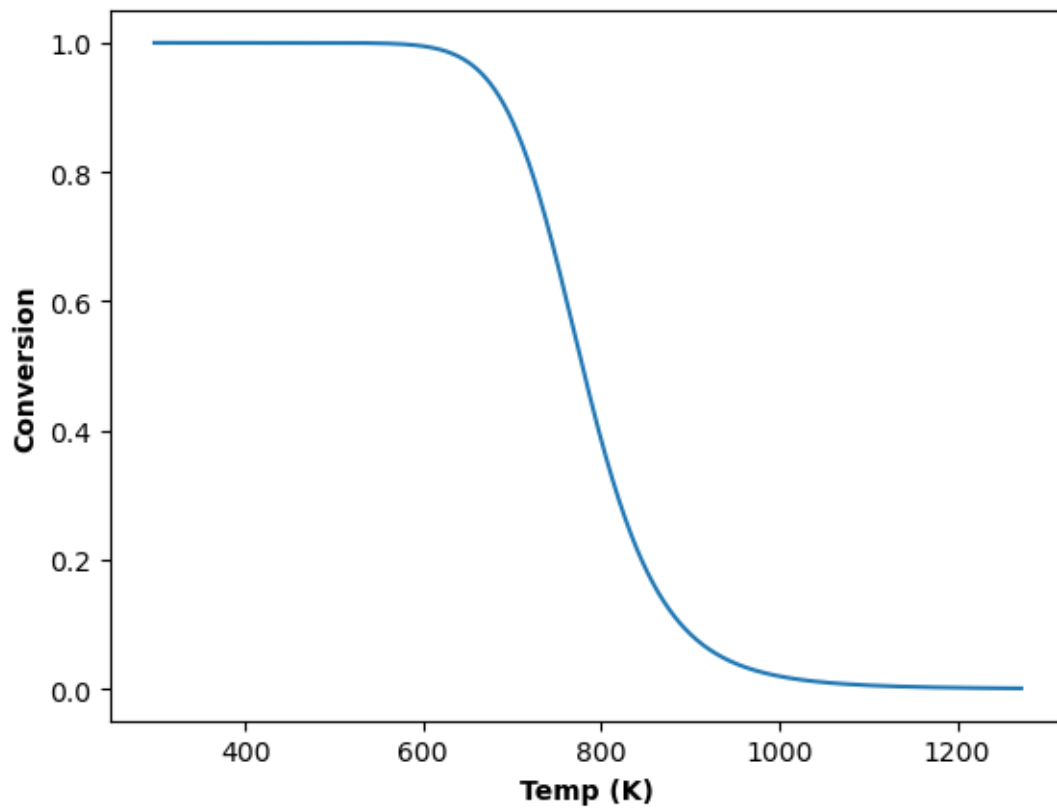
$$X_c = \frac{K(T)}{1 + K(T)}$$

where

$$K(T) = \exp\left(-\frac{\Delta H - T\Delta S}{RT}\right)$$

```
[20]: R = 8.314 # J/molK
      H = -114.14 # kJ/mol
      S = -0.146538 # kJ/mol
      temp = np.arange(298,1273)
      Kp_t = np.exp(-(H*1000-temp*S*1000)/(R*temp))
      xc = Kp_t/(1+Kp_t)

      plt.plot(temp,xc)
      plt.xlabel('Temp (K)',weight='bold')
      plt.ylabel('Conversion',weight='bold')
      plt.show()
```



```
[ ]:
```